Technical Document #450: Software Engineering - Comprehensive Overview

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Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Test-driven development (TDD) writes tests before writing code.
- Continuous deployment (CD) automatically deploys code changes to production.
- API design defines how software components communicate.